

DNS Lab

```
Command Prompt
Microsoft Windows [Version 10.0.26200.8655]
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C:\Users\bhavd>nslookup www.yahoo.co.jp
Server:  dns.cp.net.rogers.com
Address:  2607:f798:18:10:0:640:7125:5204

Non-authoritative answer:
Name:     edge12.g.yimg.jp
Address:  182.22.16.123
Aliases:  www.yahoo.co.jp

C:\Users\bhavd>nslookup -type=NS ox.ac.uk
Server:  dns.cp.net.rogers.com
Address:  2607:f798:18:10:0:640:7125:5204

Non-authoritative answer:
ox.ac.uk      nameserver = auth5.dns.ox.ac.uk
ox.ac.uk      nameserver = auth6.dns.ox.ac.uk
ox.ac.uk      nameserver = auth4.dns.ox.ac.uk
ox.ac.uk      nameserver = dns0.ox.ac.uk
ox.ac.uk      nameserver = dns2.ox.ac.uk
ox.ac.uk      nameserver = dns1.ox.ac.uk

auth5.dns.ox.ac.uk      AAAA IPv6 address = 2a00:1098:0:80:1000::10
auth6.dns.ox.ac.uk      AAAA IPv6 address = 2a02:2770:11:0:21a:4aff:febe:759b
auth4.dns.ox.ac.uk      AAAA IPv6 address = 2600:3c00:e000:19::1
auth5.dns.ox.ac.uk      internet address = 93.93.128.67
auth6.dns.ox.ac.uk      internet address = 185.24.221.32
dns2.ox.ac.uk   internet address = 163.1.2.190
dns0.ox.ac.uk   internet address = 129.67.1.190
auth4.dns.ox.ac.uk   internet address = 45.33.127.156
dns1.ox.ac.uk   internet address = 129.67.1.191

C:\Users\bhavd>nslookup -type=MX yahoo.com ns0.ox.ac.uk
** Can't find server address for 'ns0.ox.ac.uk':
Server:  dns.cp.net.rogers.com
Address:  2607:f798:18:10:0:640:7125:5204

Non-authoritative answer:
yahoo.com      MX preference = 1, mail exchanger = mta6.am0.yahoodns.net
yahoo.com      MX preference = 1, mail exchanger = mta5.am0.yahoodns.net
yahoo.com      MX preference = 1, mail exchanger = mta7.am0.yahoodns.net

C:\Users\bhavd>
```

1. IP address of www.yahoo.co.jp (Asian web server) is 182.22.16.123, resolved via local DNS server dns.cp.net.rogers.com
2. Authoritative DNS servers for Oxford University (ox.ac.uk) are:
 - 2.1. auth4.dns.ox.ac.uk (45.33.127.156)
 - 2.2. auth5.dns.ox.ac.uk (93.93.128.67)
 - 2.3. auth6.dns.ox.ac.uk (185.24.221.32)
 - 2.4. dns0.ox.ac.uk (129.67.1.190)
 - 2.5. dns1.ox.ac.uk (129.67.1.191)

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- 2.6. dns2.ox.ac.uk (163.1.2.190)
3. Yahoo mail servers (MX records) are:
 - 3.1. mta5.am0.yahoodns.net (preference 1)
 - 3.2. mta6.am0.yahoodns.net (preference 1)
 - 3.3. mta7.am0.yahoodns.net (preference 1)All have equal priority (MX preference = 1).

```
Command Prompt
C:\Users\bhavd>ipconfig /all

Windows IP Configuration

Host Name . . . . . : Bhav_Laptop
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No

Unknown adapter OpenVPN TAP-Windows6:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : TAP-Windows Adapter V9
Physical Address. . . . . : 00-FF-CB-C2-26-6E
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Unknown adapter OpenVPN Data Channel Offload:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : OpenVPN Data Channel Offload
Physical Address. . . . . :
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Wi-Fi 4:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : MediaTek Wi-Fi 7 MT7925 Wireless LAN Card #4
Physical Address. . . . . : 4A-45-E6-8B-D6-FF
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Wi-Fi 5:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : MediaTek Wi-Fi 7 MT7925 Wireless LAN Card #5
Physical Address. . . . . : 4A-45-E6-8B-C6-EF
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
```

```
Command Prompt

Section . . . . . : Answer
CNAME Record . . . . : a1992.dscd.akamai.net

Record Name . . . . . : a1992.dscd.akamai.net
Record Type . . . . . : 1
Time To Live . . . . . : 13
Data Length . . . . . : 4
Section . . . . . : Answer
A (Host) Record . . . . : 72.136.195.83

ipv6.msftconnecttest.com
-----
Record Name . . . . . : ipv6.msftconnecttest.com
Record Type . . . . . : 5
Time To Live . . . . . : 7
Data Length . . . . . : 8
Section . . . . . : Answer
CNAME Record . . . . . : ipv6.msftconnecttest.com.edgesuite.net

Record Name . . . . . : ipv6.msftconnecttest.com.edgesuite.net
Record Type . . . . . : 5
Time To Live . . . . . : 7
Data Length . . . . . : 8
Section . . . . . : Answer
CNAME Record . . . . . : a1968.i6g1.akamai.net

Record Name . . . . . : a1968.i6g1.akamai.net
Record Type . . . . . : 28
Time To Live . . . . . : 7
Data Length . . . . . : 16
Section . . . . . : Answer
AAAA Record . . . . . : 2607:f798:d02:2::4888:c21a

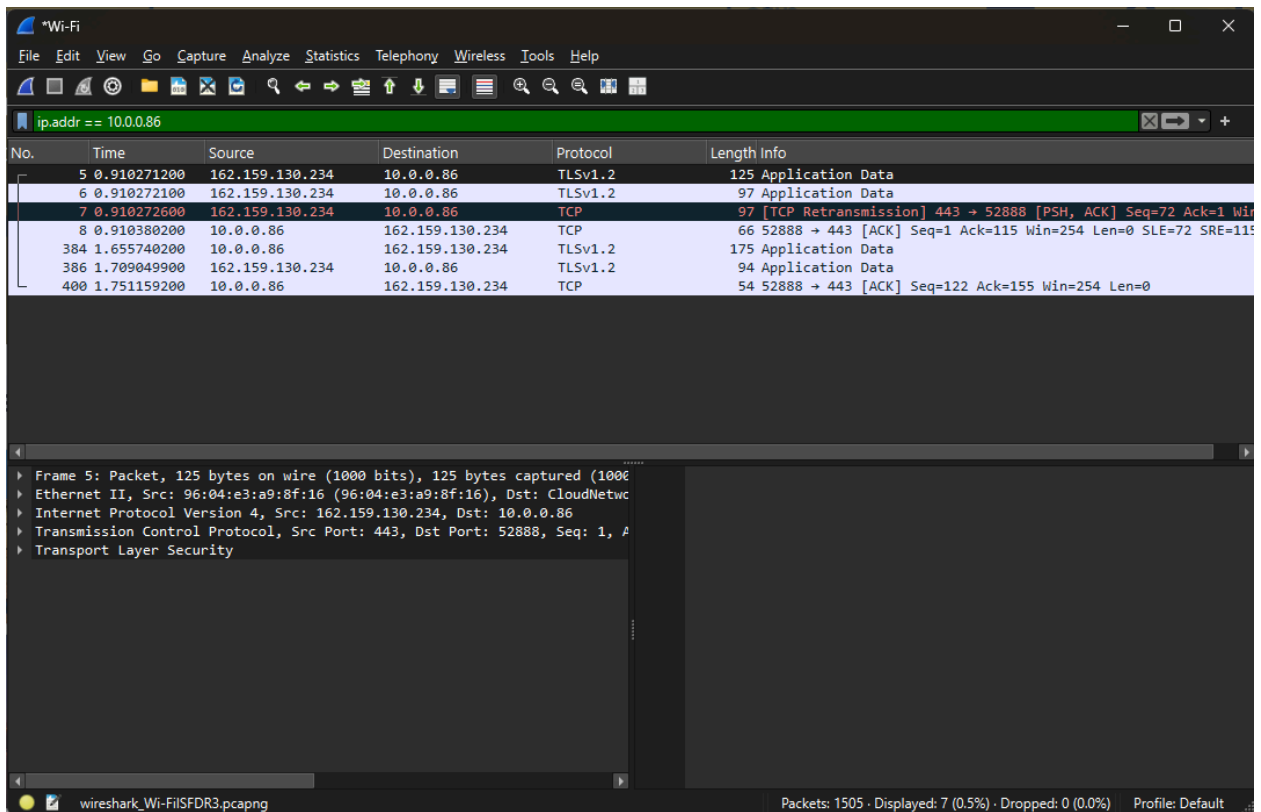
C:\Users\bhavd>ipconfig /flushdns

Windows IP Configuration

Successfully flushed the DNS Resolver Cache.

C:\Users\bhavd>
```

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Wireshark capture showing TLS traffic. The filter is `ip.addr == 10.0.0.86`. The packet list shows several packets, including a TLSv1.2 application data packet (No. 5) and a TCP retransmission (No. 7). The packet details pane shows the structure of the selected packet (No. 5), including Ethernet II, Internet Protocol Version 4, Transmission Control Protocol, and Transport Layer Security.

No.	Time	Source	Destination	Protocol	Length	Info
5	0.910271200	162.159.130.234	10.0.0.86	TLSv1.2	125	Application Data
6	0.910272100	162.159.130.234	10.0.0.86	TLSv1.2	97	Application Data
7	0.910272600	162.159.130.234	10.0.0.86	TCP	97	[TCP Retransmission] 443 → 52888 [PSH, ACK] Seq=72 Ack=1 Win=0 Len=0
8	0.910380200	10.0.0.86	162.159.130.234	TCP	66	52888 → 443 [ACK] Seq=1 Ack=115 Win=254 Len=0 SLE=72 SRE=115
384	1.655740200	10.0.0.86	162.159.130.234	TLSv1.2	175	Application Data
386	1.709049900	162.159.130.234	10.0.0.86	TLSv1.2	94	Application Data
400	1.751159200	10.0.0.86	162.159.130.234	TCP	54	52888 → 443 [ACK] Seq=122 Ack=155 Win=254 Len=0

Frame 5: Packet, 125 bytes on wire (1000 bits), 125 bytes captured (1000 bits) on interface 0

Ethernet II, Src: 96:04:e3:a9:8f:16 (96:04:e3:a9:8f:16), Dst: CloudNetwork

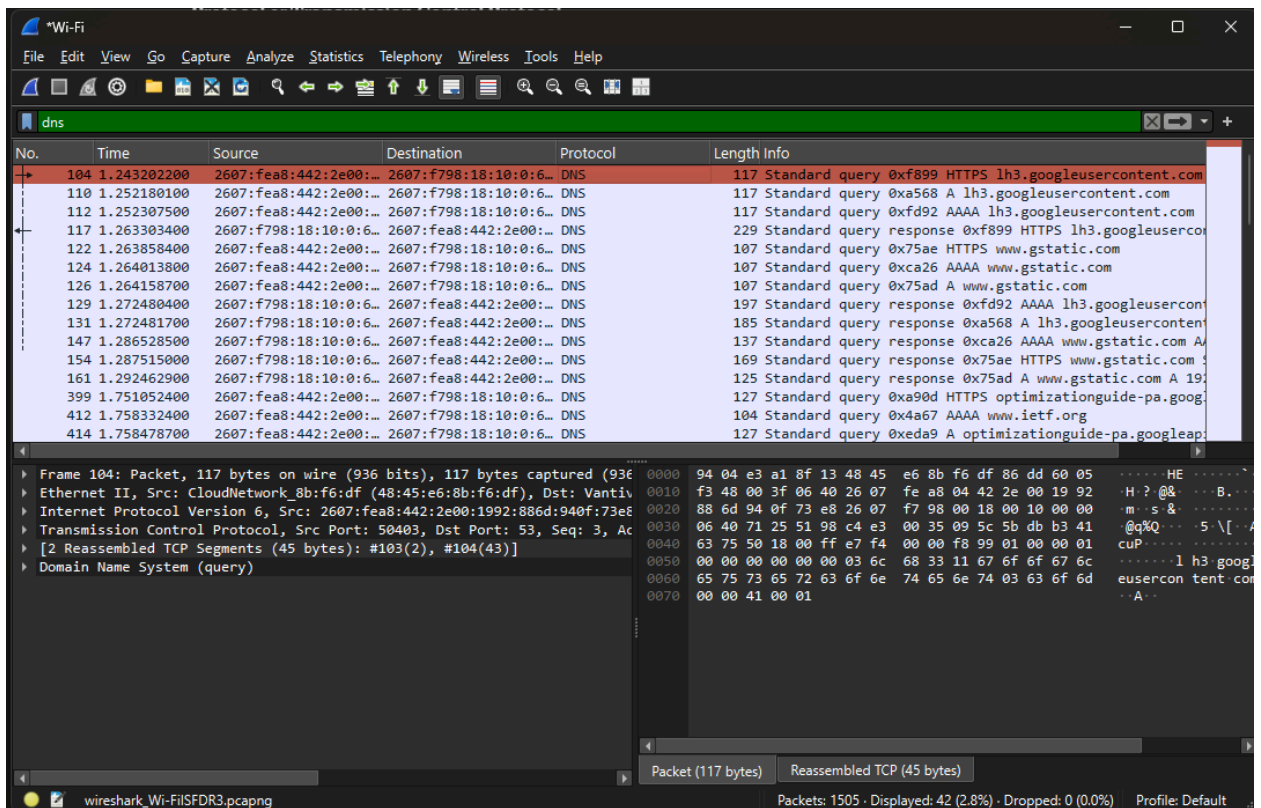
Internet Protocol Version 4, Src: 162.159.130.234, Dst: 10.0.0.86

Transmission Control Protocol, Src Port: 443, Dst Port: 52888, Seq: 1, Len: 0

Transport Layer Security

wireshark_Wi-FiISFDR3.pcapng

Packets: 1505 · Displayed: 7 (0.5%) · Dropped: 0 (0.0%) · Profile: Default



Wireshark capture showing DNS traffic. The filter is `dns`. The packet list shows several DNS queries and responses. The packet details pane shows the structure of the selected packet (No. 104), including Ethernet II, Internet Protocol Version 6, Transmission Control Protocol, and Domain Name System (query).

No.	Time	Source	Destination	Protocol	Length	Info
104	1.243202200	2607:fea8:442:2e00::...	2607:f798:18:10:0:6...	DNS	117	Standard query 0xf899 HTTPS lh3.googleusercontent.com
110	1.252180100	2607:fea8:442:2e00::...	2607:f798:18:10:0:6...	DNS	117	Standard query 0xa568 A lh3.googleusercontent.com
112	1.252307500	2607:fea8:442:2e00::...	2607:f798:18:10:0:6...	DNS	117	Standard query 0xfd92 AAAA lh3.googleusercontent.com
117	1.263303400	2607:f798:18:10:0:6...	2607:fea8:442:2e00::...	DNS	229	Standard query response 0xf899 HTTPS lh3.googleusercontent.com
122	1.263858400	2607:fea8:442:2e00::...	2607:f798:18:10:0:6...	DNS	107	Standard query 0x75ae HTTPS www.gstatic.com
124	1.264013800	2607:fea8:442:2e00::...	2607:f798:18:10:0:6...	DNS	107	Standard query 0xca26 AAAA www.gstatic.com
126	1.264158700	2607:fea8:442:2e00::...	2607:f798:18:10:0:6...	DNS	107	Standard query 0x75ad A www.gstatic.com
129	1.272480400	2607:f798:18:10:0:6...	2607:fea8:442:2e00::...	DNS	197	Standard query response 0xfd92 AAAA lh3.googleusercontent.com
131	1.272481700	2607:f798:18:10:0:6...	2607:fea8:442:2e00::...	DNS	185	Standard query response 0xa568 A lh3.googleusercontent.com
147	1.286528500	2607:f798:18:10:0:6...	2607:fea8:442:2e00::...	DNS	137	Standard query response 0xca26 AAAA www.gstatic.com
154	1.287515000	2607:f798:18:10:0:6...	2607:fea8:442:2e00::...	DNS	169	Standard query response 0x75ae HTTPS www.gstatic.com
161	1.292462900	2607:f798:18:10:0:6...	2607:fea8:442:2e00::...	DNS	125	Standard query response 0x75ad A www.gstatic.com
399	1.751052400	2607:fea8:442:2e00::...	2607:f798:18:10:0:6...	DNS	127	Standard query 0xa90d HTTPS optimizationguide-pa.googleap
412	1.758332400	2607:fea8:442:2e00::...	2607:f798:18:10:0:6...	DNS	104	Standard query 0xa67 AAAA www.ietf.org
414	1.758478700	2607:fea8:442:2e00::...	2607:f798:18:10:0:6...	DNS	127	Standard query 0xeda9 A optimizationguide-pa.googleap

Frame 104: Packet, 117 bytes on wire (936 bits), 117 bytes captured (936 bits) on interface 0

Ethernet II, Src: CloudNetwork_8b:f6:df (48:45:e6:8b:f6:df), Dst: Vantiv

Internet Protocol Version 6, Src: 2607:fea8:442:2e00::1992:886d:940f:73e8, Dst: 2607:f798:18:10:0:6::1

Transmission Control Protocol, Src Port: 50403, Dst Port: 53, Seq: 3, Len: 45

[2 Reassembled TCP Segments (45 bytes): #103(2), #104(43)]

Domain Name System (query)

Packet (117 bytes) Reassembled TCP (45 bytes)

wireshark_Wi-FiISFDR3.pcapng

Packets: 1505 · Displayed: 42 (2.8%) · Dropped: 0 (0.0%) · Profile: Default

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4. DNS messages here are sent over TCP (though DNS typically uses UDP port 53; TCP is used for larger responses or when UDP fails).
5. Destination port of DNS query: 53 & Source port of DNS response: 53
6. The DNS query is sent to 2607:f798:18:10:0:640:7125:5198

The screenshot shows a Wireshark capture of DNS traffic. The packet list at the top shows a series of DNS queries and responses. The selected packet (No. 104) is a DNS query from 2607:fea8:442:2e00:: to 2607:f798:18:10:0:640:7125:5198. The packet details pane shows the query structure, including the transaction ID (0xf899) and the query type (Standard query). The packet bytes pane shows the raw data of the query, including the domain name (lh3.googleusercontent.com) and the query type (0x0000).

No.	Time	Source	Destination	Protocol	Length	Info
104	1.243202200	2607:fea8:442:2e00::	2607:f798:18:10:0:640:7125:5198	DNS	117	Standard query 0xf899 HTTPS lh3.googleusercontent.com
110	1.252180100	2607:fea8:442:2e00::	2607:f798:18:10:0:640:7125:5198	DNS	117	Standard query 0xa568 A lh3.googleusercontent.com
112	1.252307500	2607:fea8:442:2e00::	2607:f798:18:10:0:640:7125:5198	DNS	117	Standard query 0xfd92 AAAA lh3.googleusercontent.com
117	1.263303400	2607:f798:18:10:0:640:7125:5198	2607:fea8:442:2e00::	DNS	229	Standard query response 0xf899 HTTPS lh3.googleusercontent.com
122	1.263858400	2607:fea8:442:2e00::	2607:f798:18:10:0:640:7125:5198	DNS	107	Standard query 0x75ae HTTPS www.gstatic.com
124	1.264013800	2607:fea8:442:2e00::	2607:f798:18:10:0:640:7125:5198	DNS	107	Standard query 0xca26 AAAA www.gstatic.com
126	1.264158700	2607:fea8:442:2e00::	2607:f798:18:10:0:640:7125:5198	DNS	107	Standard query 0x75ad A www.gstatic.com
129	1.272480400	2607:f798:18:10:0:640:7125:5198	2607:fea8:442:2e00::	DNS	197	Standard query response 0xfd92 AAAA lh3.googleusercontent.com
131	1.272481700	2607:f798:18:10:0:640:7125:5198	2607:fea8:442:2e00::	DNS	185	Standard query response 0xa568 A lh3.googleusercontent.com
147	1.286528500	2607:f798:18:10:0:640:7125:5198	2607:fea8:442:2e00::	DNS	137	Standard query response 0xca26 AAAA www.gstatic.com
154	1.287515000	2607:f798:18:10:0:640:7125:5198	2607:fea8:442:2e00::	DNS	169	Standard query response 0x75ae HTTPS www.gstatic.com
161	1.292462900	2607:f798:18:10:0:640:7125:5198	2607:fea8:442:2e00::	DNS	125	Standard query response 0x75ad A www.gstatic.com
399	1.751052400	2607:fea8:442:2e00::	2607:f798:18:10:0:640:7125:5198	DNS	127	Standard query 0xa90d HTTPS optimizationguide-pa.googleusercontent.com
412	1.758332400	2607:fea8:442:2e00::	2607:f798:18:10:0:640:7125:5198	DNS	104	Standard query 0x4a67 AAAA www.ietf.org
414	1.758478700	2607:fea8:442:2e00::	2607:f798:18:10:0:640:7125:5198	DNS	127	Standard query 0xeda9 A optimizationguide-pa.googleusercontent.com

Packet (117 bytes) Reassembled TCP (45 bytes)

Number of answers in packet (dns.count.answers), 2 bytes

Packets: 1505 · Displayed: 42 (2.8%) · Dropped: 0 (0.0%) Profile: Default

7. The DNS query is of type HTTPS (as seen in the Info column). The query message contains 0 answers — this is expected, queries never contain answers, only responses do.

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The image shows a Wireshark packet capture of DNS traffic. The top pane displays a list of packets, with packet 129 selected. The bottom-left pane shows the details of the selected packet, and the bottom-right pane shows the raw packet data in hexadecimal and ASCII.

Packet List:

No.	Time	Source	Destination	Protocol	Length	Info
104	1.243202200	2607:fea8:442:2e00::...	2607:f798:18:10:0:6...	DNS	117	Standard query 0xf899 HTTPS lh3.googleusercontent.com
110	1.252180100	2607:fea8:442:2e00::...	2607:f798:18:10:0:6...	DNS	117	Standard query 0xa568 A lh3.googleusercontent.com
112	1.252307500	2607:fea8:442:2e00::...	2607:f798:18:10:0:6...	DNS	117	Standard query 0xfd92 AAAA lh3.googleusercontent.com
117	1.263303400	2607:f798:18:10:0:6...	2607:fea8:442:2e00::...	DNS	229	Standard query response 0xf899 HTTPS lh3.googleusercontent.com
122	1.263858400	2607:fea8:442:2e00::...	2607:f798:18:10:0:6...	DNS	107	Standard query 0x75ae HTTPS www.gstatic.com
124	1.264013800	2607:fea8:442:2e00::...	2607:f798:18:10:0:6...	DNS	107	Standard query 0xca26 AAAA www.gstatic.com
126	1.264158700	2607:fea8:442:2e00::...	2607:f798:18:10:0:6...	DNS	107	Standard query 0x75ad A www.gstatic.com
129	1.272480400	2607:f798:18:10:0:6...	2607:fea8:442:2e00::...	DNS	197	Standard query response 0xfd92 AAAA lh3.googleusercontent.com
131	1.272481700	2607:f798:18:10:0:6...	2607:fea8:442:2e00::...	DNS	185	Standard query response 0xa568 A lh3.googleusercontent.com
147	1.286528500	2607:f798:18:10:0:6...	2607:fea8:442:2e00::...	DNS	137	Standard query response 0xca26 AAAA www.gstatic.com
154	1.287515000	2607:f798:18:10:0:6...	2607:fea8:442:2e00::...	DNS	169	Standard query response 0x75ae HTTPS www.gstatic.com
161	1.292462900	2607:f798:18:10:0:6...	2607:fea8:442:2e00::...	DNS	125	Standard query response 0x75ad A www.gstatic.com
399	1.751052400	2607:fea8:442:2e00::...	2607:f798:18:10:0:6...	DNS	127	Standard query 0xa90d HTTPS optimizationguide-pa.google.com
412	1.758332400	2607:fea8:442:2e00::...	2607:f798:18:10:0:6...	DNS	104	Standard query 0xa67 AAAA www.ietf.org
414	1.758478700	2607:fea8:442:2e00::...	2607:f798:18:10:0:6...	DNS	127	Standard query 0xeda9 A optimizationguide-pa.google.com

Packet 129 Details:

Questions: 1
Answer RRs: 2
Authority RRs: 0
Additional RRs: 0

Answers:

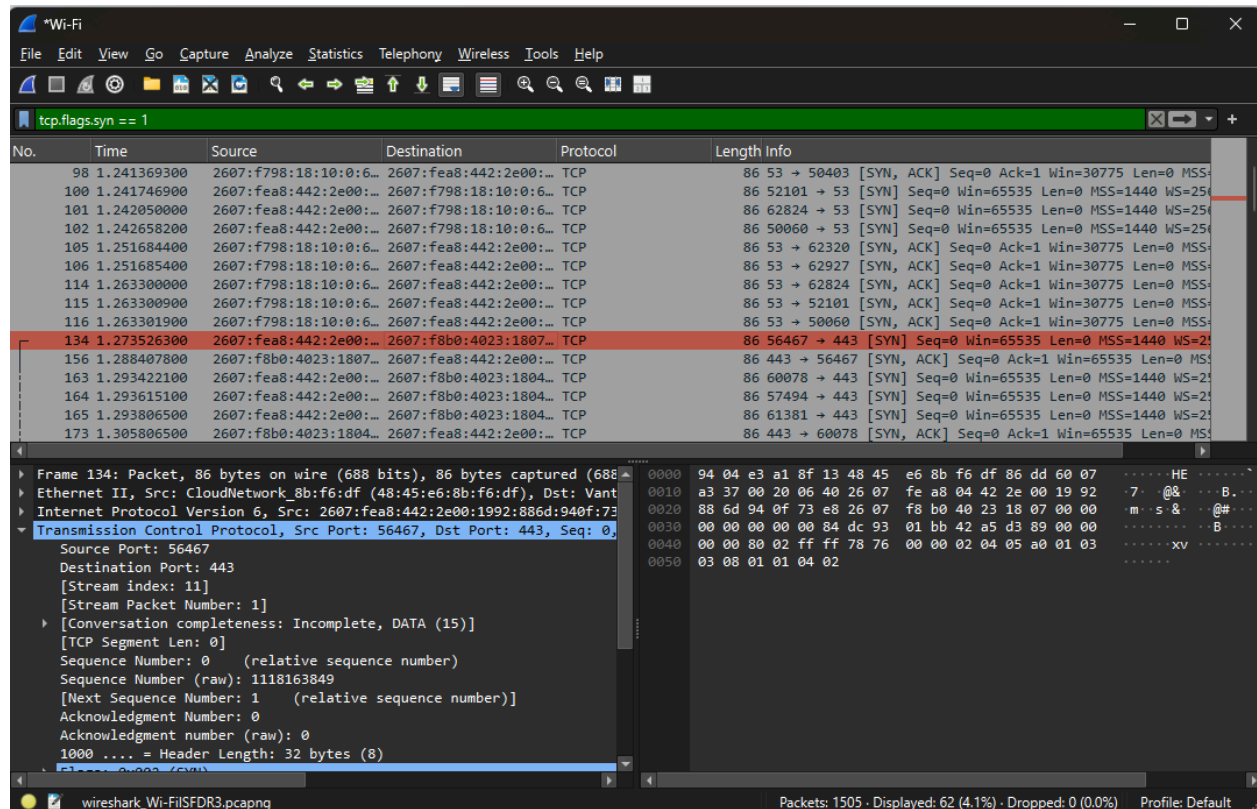
- lh3.googleusercontent.com: type CNAME, class IN, cname googlehosted.l.googleusercontent.com
Name: lh3.googleusercontent.com
Type: CNAME (5) (Canonical NAME for an alias)
Class: IN (0x0001)
Time to live: 73 (1 minute, 13 seconds)
Data length: 38
CNAME: googlehosted.l.googleusercontent.com
- googlehosted.l.googleusercontent.com: type AAAA, class IN, addr
Name: googlehosted.l.googleusercontent.com
Type: AAAA (28) (IPv6 Address)
Class: IN (0x0001)

Raw Data:

0000 48 45 e6 8b f6 df 96 04 e3 a9 8f 16 86 dd 60 10 HE.....&.....
0010 00 00 00 8f 06 38 26 07 f7 98 00 18 00 10 00 00 @g%Q&...B...
0020 06 40 71 25 51 98 26 07 fe a8 04 42 2e 00 19 92 .m's 5 ..I..ul
0030 88 6d 94 0f 73 e8 00 35 f5 cf 49 05 da 75 6c 59 ..P..X...y...
0040 c7 87 50 18 1c 20 58 c9 00 00 00 79 fd 92 81 80lh3-go
0050 00 01 00 02 00 00 00 00 03 6c 68 33 11 67 6f 6fgleuserc ontent.
0060 67 6c 65 75 73 65 72 63 6f 6e 74 65 6e 74 03 63 om.....
0070 6f 6d 00 00 1c 00 01 c0 0c 00 05 00 01 00 00 00 ..&..goog lehoste
0080 49 00 26 0c 67 6f 6f 67 6c 65 68 6f 73 74 65 64 .l.googl euserco
0090 01 6c 11 67 6f 6f 67 6c 65 75 73 65 72 63 6f 6e tent.com ..7...
00a0 74 65 6e 74 03 63 6f 6d 00 c0 37 00 1c 00 01 00 ..9..&..@#...
00b0 00 00 39 00 10 26 07 f8 b0 40 23 18 07 00 00 00
00c0 00 00 00 00 84

8. The DNS response provides 2 answers:
 - 8.1. A CNAME record — lh3.googleusercontent.com is an alias for googlehosted.l.googleusercontent.com
 - 8.2. An AAAA record (IPv6 address) — googlehosted.l.googleusercontent.com resolves to 2607:f8b0:4023:1807::84

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9. Yes, The destination IP of the SYN packet is 2607:f8b0:4023:1804::84, which matches exactly the AAAA address provided in the DNS response (packet 129). This confirms that the browser used the DNS response to know where to connect.
10. Yes, the host issues new DNS queries for additional hostnames as the page loads. For example, a query was seen for safebrowsing.google.com && optimizationguide-pa.googleapis.com, this is Chrome checking URLs against Google's safe browsing database. Additional DNS queries would also appear for any images or resources embedded in the ietf.org page, each requiring its own DNS resolution.

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The image displays a Wireshark packet capture of a DNS query and response, alongside a Windows Command Prompt window showing the results of a `nslookup` command.

Wireshark Packet Capture:

- Packet 19: Standard query request from 2607:f798:18:10:0:640:7125:5204 to 2607:f798:18:10:0:640:7125:5204. Protocol: DNS. Length: 152.
- Packet 20: Standard query response from 2607:f798:18:10:0:640:7125:5204 to 2607:f798:18:10:0:640:7125:5204. Protocol: DNS. Length: 223.

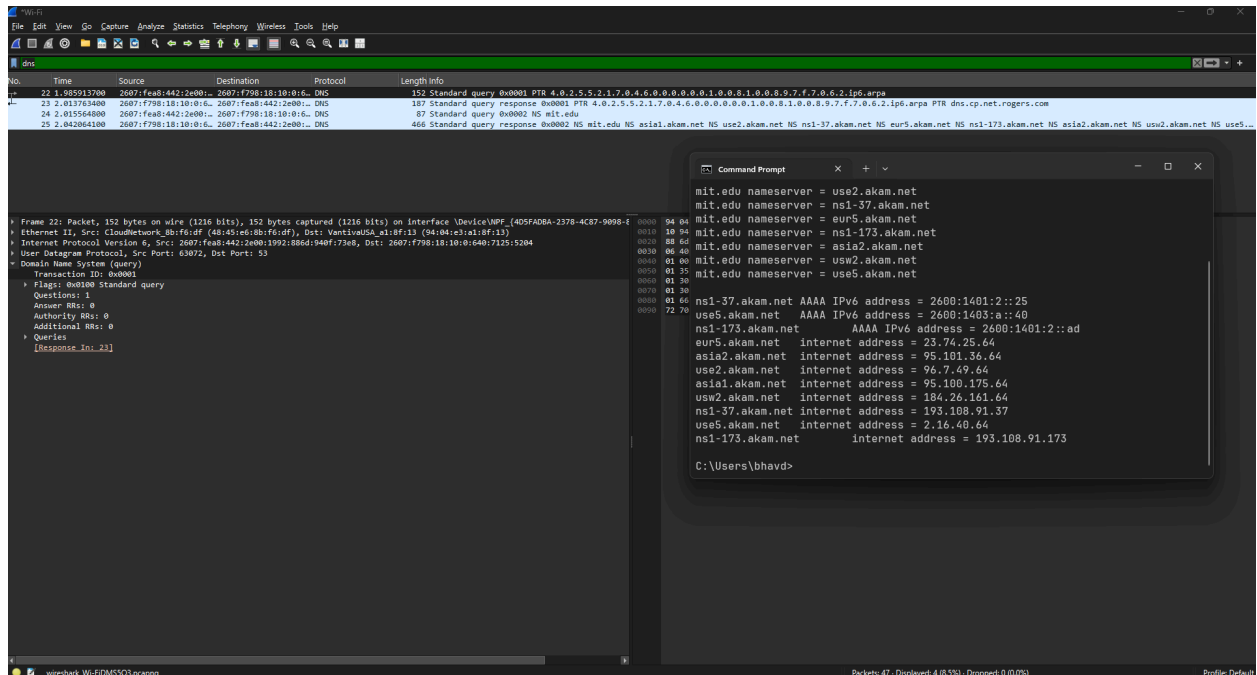
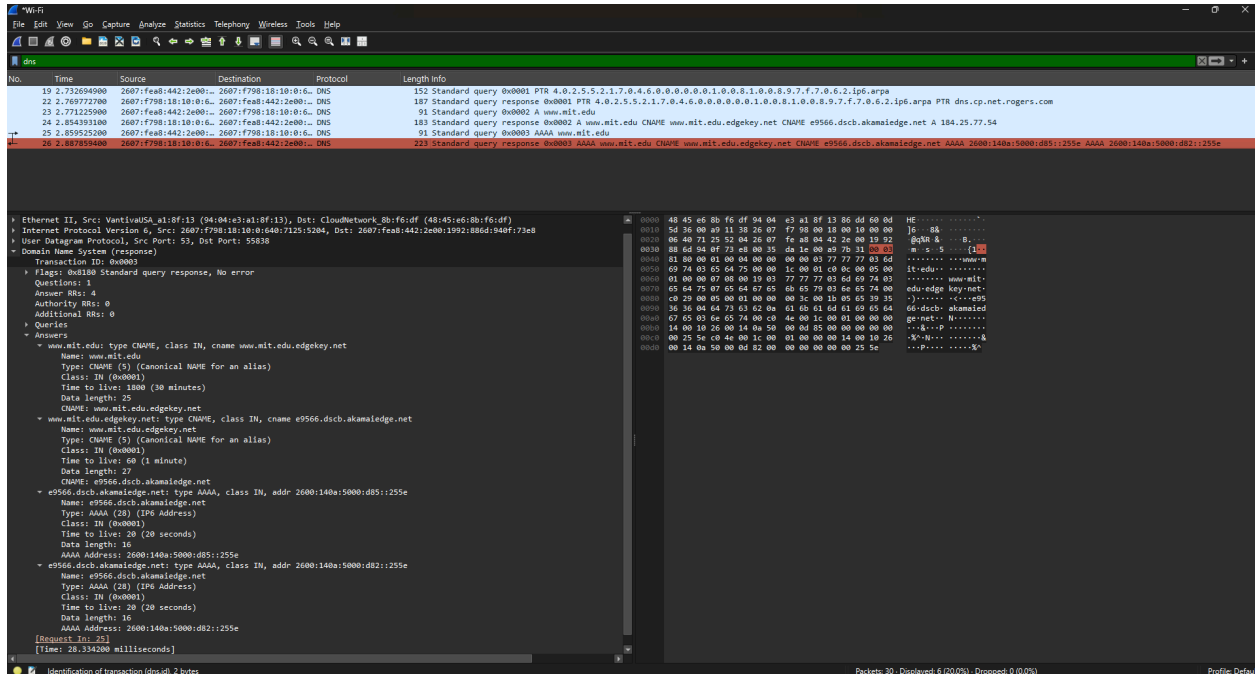
Command Prompt Output:

```
C:\Users\bhavd>nslookup www.mit.edu
Server: dns.cp.net.rogers.com
Address: 2607:f798:18:10:0:640:7125:5204

Non-authoritative answer:
Name: e9566.dscb.akamaiedge.net
Addresses: 2600:140a:5000:d85::255e
2600:140a:5000:d82::255e
184.25.77.54
Aliases: www.mit.edu
www.mit.edu.edgekey.net
```

11. Destination port of DNS query: 53 && Source port of DNS response: 53 (same as before, DNS always uses port 53)
12. DNS query sent to 2607:f798:18:10:0:640:7125:5204, yes this is my default local DNS server (dns.cp.net.rogers.com) confirmed by ipconfig
13. The DNS query is of type A (Host address lookup) for www.mit.edu. The query contains 0 answers; queries never contain answers.
14. The DNS response provides 4 answers:
 - 14.1. CNAME — www.mit.edu is an alias for www.mit.edu.edgekey.net
 - 14.2. CNAME — www.mit.edu.edgekey.net is an alias for e9566.dscb.akamaiedge.net (Akamai CDN)
 - 14.3. AAAA — e9566.dscb.akamaiedge.net resolves to 2600:140a:5000:d85::255e
 - 14.4. AAAA — e9566.dscb.akamaiedge.net also resolves to 2600:140a:5000:d82::255e

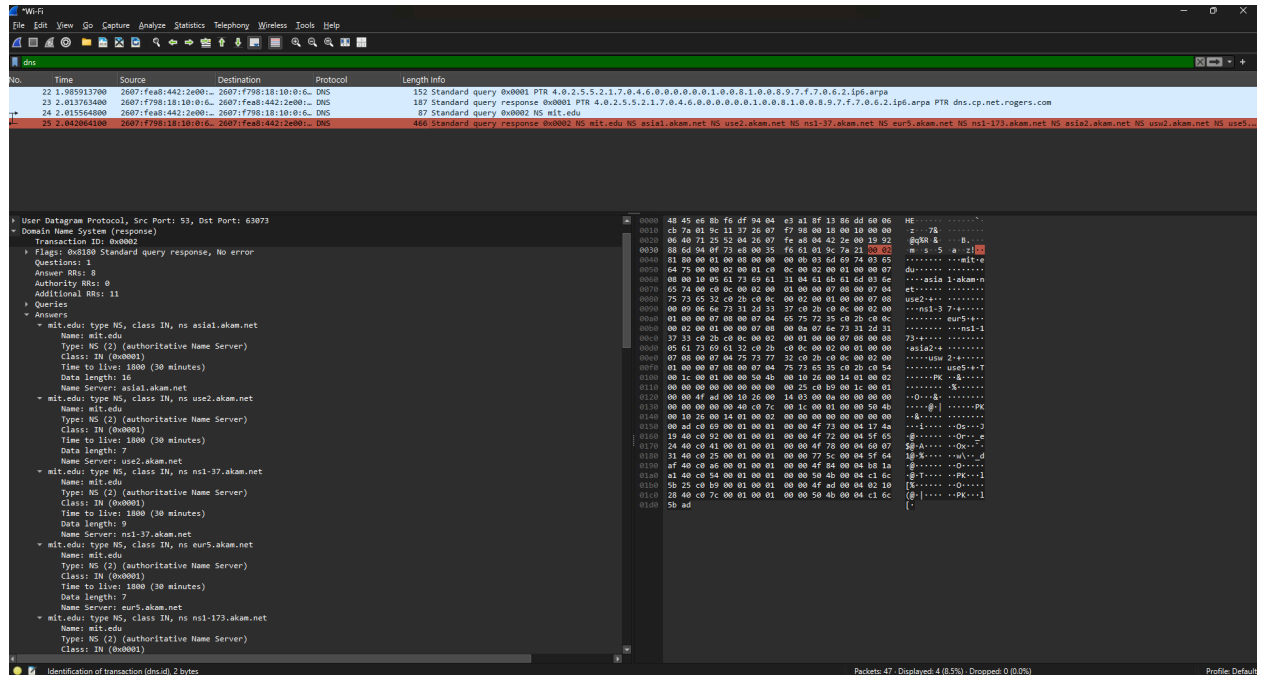
DNS Lab



16. DNS query sent to 2607:f798:18:10:0:640:7125:5204 — yes, this is your default local DNS server (dns.cp.net.rogers.com)
17. The DNS query is of type NS (Name Server lookup) for mit.edu. The query contains 0 answers, as always, queries never contain answers.

18. The response provides these MIT nameservers:

- 18.1. asia1.akam.net (95.100.175.64)
- 18.2. asia2.akam.net (95.101.36.64)
- 18.3. eur5.akam.net (23.74.25.64)
- 18.4. use2.akam.net (96.7.49.64)
- 18.5. use5.akam.net (2.16.40.64)
- 18.6. usw2.akam.net (184.26.161.64)
- 18.7. ns1-37.akam.net (193.108.91.37)
- 18.8. ns1-173.akam.net (193.108.91.173)



19.

DNS Lab

The screenshot displays a Wireshark capture of DNS traffic and a corresponding Command Prompt window. The Wireshark interface shows a list of captured packets, with the selected packet (No. 34) expanded to show its details. The details pane highlights the 'Domain Name System (Query)' section, indicating a standard query for 'www.aiit.or.kr'.

The Command Prompt window shows the execution of the command `C:\Users\bhavd>nslookup www.aiit.or.kr bitsy.mit.edu`. The output indicates that the DNS request timed out, and the server is unknown, with the address 18.0.72.3 displayed.

Packet 34 details (Domain Name System (Query)):

- Transaction ID: 0x0036
- Flags: 0x0000 Standard query
- Questions: 1
- Answer RRs: 0
- Authority RRs: 0
- Additional RRs: 0
- Queries

Command Prompt output:

```
C:\Users\bhavd>nslookup www.aiit.or.kr bitsy.mit.edu
DNS request timed out.
    timeout was 2 seconds.
Server: Unknown
Address: 18.0.72.3

DNS request timed out.
    timeout was 2 seconds.
DNS request timed out.
    timeout was 2 seconds.
DNS request timed out.
    timeout was 2 seconds.
DNS request timed out.
    timeout was 2 seconds.
DNS request timed out.
    timeout was 2 seconds.
** Request to Unknown timed-out

C:\Users\bhavd>
```

20. The DNS query was sent to 18.0.72.3 — this is the IP address of bitsy.mit.edu, NOT your default local DNS server (which is dns.cp.net.rogers.com). The query was sent directly to MIT's nameserver as specified in the nslookup command.
21. The DNS query is of type **A** (Host address lookup) for www.aiit.or.kr. The query contains **0 answers**.
22. No answers were provided — the request timed out because bitsy.mit.edu is no longer responding. In the original lab, the expected answer would have been 1 answer containing the A record (IPv4 address) for www.aiit.or.kr.

DNS Lab

The image shows a Wireshark capture of DNS traffic. The top pane displays a list of captured packets, with packet 40 selected. The bottom pane shows the detailed view of the selected packet, which is a DNS response from 2607:f798:18:10:0:6... to 2607:fea8:442:2e00:...

Packet List:

No.	Time	Source	Destination	Protocol	Length	Info
34	3.131230800	2607:f798:18:10:0:6...	2607:fea8:442:2e00:...	DNS	109	Standard query response 0xcd36 A bitsy.mit.edu A 18.0
35	3.131460100	2607:fea8:442:2e00:...	2607:f798:18:10:0:6...	DNS	93	Standard query 0x82a0 AAAA bitsy.mit.edu
36	3.131467100	2607:fea8:442:2e00:...	2607:f798:18:10:0:6...	DNS	93	Standard query 0xcd36 A bitsy.mit.edu
38	3.141487000	2607:f798:18:10:0:6...	2607:fea8:442:2e00:...	DNS	158	Standard query response 0x82a0 AAAA bitsy.mit.edu SOA
39	3.144446800	10.0.0.86	18.0.72.3	DNS	82	Standard query 0x0001 PTR 3.72.0.18.in-addr.arpa
40	3.153643600	2607:f798:18:10:0:6...	2607:fea8:442:2e00:...	DNS	158	Standard query response 0x82a0 AAAA bitsy.mit.edu SOA

Domain Name System (response)
Transaction ID: 0x82a0
Flags: 0x8180 Standard query response, No error
Questions: 1
Answer RRs: 0
Authority RRs: 1
Additional RRs: 0
Queries:
Authoritative nameservers:
mit.edu: type SOA, class IN, mname use2.akam.net
Name: mit.edu
Type: SOA (6) (Start Of a zone of Authority)
Class: IN (0x0001)
Time to live: 1800 (30 minutes)
Data length: 53
Primary name server: use2.akam.net
Responsible authority's mailbox: network-request.mit.edu
[Responsible authority's mailbox name: network-request@mit.edu]
Serial Number: 23046
Refresh Interval: 900 (15 minutes)
Retry Interval: 900 (15 minutes)
Expire limit: 3600000 (41 days, 16 hours)
Minimum TTL: 3600 (1 hour)
[Request In: 35]
[Time: 22.183500 milliseconds]

Packet Details:

0000 48 45 e6 8b f6 df 94 04 e3 a1 8f 13 86 dd 60 0b HE
0010 c8 59 00 68 11 38 26 07 f7 98 00 18 00 10 00 00 .Y.h.8&
0020 06 40 71 25 51 98 26 07 fe a8 04 42 2e 00 19 92 @%Q & ...B...
0030 88 6d 94 0f 73 e8 00 35 c4 95 00 68 91 75 82 a0 .m.s.-5...h.u...
0040 81 80 00 01 00 00 00 01 00 00 05 62 69 74 73 79bitsy...
0050 03 6d 69 74 03 65 64 75 00 00 1c 00 01 c0 12 00 .mit.edu.....
0060 06 00 01 00 00 07 08 00 35 04 75 73 65 32 04 615-use2.a...
0070 6b 61 6d 03 6e 65 74 00 0f 6e 65 74 77 6f 72 6b kam.net..network
0080 2d 72 65 71 75 65 73 74 c0 12 00 00 5a 06 00 00 -request....Z...
0090 03 84 00 00 03 84 00 36 ee 80 00 00 0e 106.....

Status Bar: Identification of transaction (dns.id), 2 bytes | Packets: 176 - Displayed: 13 (7.4%) - Dropped: 0 (0.0%) | Profile: Default

23.